Week 8 — Final Engineering Test Findings & Validation Report

University Student-Support Case Agent

Course	BSE4104 Emerging Trends in Software Engineering
Project	University Student-Support Case Agent
Institution	Makerere University — College of Computing and Information Sciences
Date	23 October 2026
Quality & Security Lead	Garanga John (2300722975 | 23/X/22975/PS)
Reviewed by	Kabanda Arthur (Project/Requirements Lead)

1. Executive Test Summary

As part of the Week 8 final consolidation and hardening phase, the entire verification suite was executed to validate deterministic tool boundaries, bounded orchestrator loop behavior, prompt guardrails, session/state contracts, MCP interoperability, and retrieval-augmented generation (RAG) components.

```
============================= Test Session Summary ==============================
Platform: Windows-11 / Python 3.12.x / pytest 8.x
Root Directory: d:\capstone_project\BSE4104
Total Test Items Collected: 502
Passed: 496
Skipped: 6 (Opt-in live network/external model tests requiring production quota)
Warnings: 1 (Starlette TestClient deprecated portal alias warning)
Execution Time: 19.53 seconds
Regression Status: ZERO REGRESSIONS DETECTED
================================================================================
```

2. Test Suite Breakdown by Architectural Layer

Layer / Subsystem	Test Module	Passed	Skipped	Primary Verification Objective
Health & Sanity	tests/test_health.py	6	0	FastAPI /health endpoint, readiness probes, environment verification.
Foundation Model	tests/test_model_client.py	14	0	Provider configuration, schema conversion, fallback handling, tool declarations.
Retrieval & RAG	tests/test_retrieval.py, tests/test_rag_evaluation.py	42	0	Chroma vector index queries, provenance metadata integrity, similarity scoring, top-k retrieval bounds.
Deterministic Tools	tests/test_tools.py	118	0	get_case_status and create_support_ticket validation, regex formatting, mock database persistence, parameter typing.
Human-in-the-Loop	tests/test_guardrails.py	58	0	Examination disputes and HIGH urgency escalation gates (PENDING_STAFF_APPROVAL), disallowed outcome alteration blocking (DISALLOWED_TOPIC).

Layer / Subsystem	Test Module	Passed	Skipped	Primary Verification Objective
Bounded Orchestrator	tests/test_orchestrator.py, tests/test_stop_conditions.py	86	0	Sense/Plan/Act/Observe/Stop cycle, MAX_TOOL_ROUNDS=4 enforcement, repeated tool-call refusal, four canonical stop reasons.
Session & State	tests/test_session_store.py, tests/test_state_contract.py, tests/test_session_chat.py	64	0	JSON session persistence, atomic replacement, corruption quarantine, 24-hr TTL expiration, session isolation.
MCP Integration	tests/test_mcp_adapter.py	38	0	JSON-RPC 2.0 stdio protocol handshake, tool catalogue derivation, input validation, error code translation parity.
Observability & Traces	tests/test_observability.py, tests/test_week7_capture.py	40	0	Turn trace construction, stage timing, retry accounting, sanitized log emission.
Evaluation Benchmarks	tests/test_week7_evaluation.py, tests/test_week7_guardrails.py, tests/test_evaluation.py	30	6	30 synthetic evaluation cases (Normal, Edge, Misleading, Adversarial, Tool Failure, Unauthorized Action). Skipped items require live model quota.
TOTAL	**18 Test Modules**	**496**	**6**	**Comprehensive Full-System Automated Verification**

3. Retested Failure Catalogue Items (Post-Fix Verification)

All 6 high-priority vulnerabilities identified in the Week 7 failure catalogue were retested under automated regression suites and confirmed fully resolved:

1. **Adversarial Outcome Rewording (G1)**

   Issue: Demands to fix grades or clear fee balances using varied vocabulary (fix, scrub, zero, sort out) previously evaded simple substring matchers.

   Fix: Regex-based demand verb and protected subject detection in src/tools/handlers.py.

   Verification: test_g1_reworded_outcome_demands_are_refused passed across all 18 adversarial variants (HTTP 403 DISALLOWED_TOPIC).

2. **Volunteered Credentials in Free Text (G2)**

   Issue: Students entering passwords, PINs, or banking details into ticket descriptions resulted in cleartext storage.

   Fix: Deterministic secret redaction routine _redact_secrets() applied at the tool boundary prior to validation or persistence.

   Verification: test_g2_credentials_are_never_persisted passed; masked values ([REDACTED_SECRET]) stored in mock database.

3. **Unicode / Non-ASCII Identifier Injection (G3)**

   Issue: Full-width Unicode numbers (e.g., CAS-２０２６-００１) passed \d regex and failed silently as CASE_NOT_FOUND.

   Fix: Strict ASCII [0-9] regular expressions enforced across all ID validators.

   Verification: test_g3_non_ascii_digits_are_invalid_not_not_found passed (HTTP 400 INVALID_ID_FORMAT).

4. **Double Retrieval Failure Stalling (G4)**

   Issue: Successive tool errors risked indefinite agent replanning loops.

Fix: Strike accounting in src/orchestrator.py halts execution after 2 consecutive failures with explicit tool_error stop reason.

Verification: test_tool_failure_replans_once_then_stops passed.

5. **Transient Model API Faults (G5)**

   Issue: Intermittent network drops or rate-limit spikes crashed single-turn interactions.

   Fix: Exponential backoff retry handler inside src/model_client.py preserving audit trace attempts.

   Verification: test_retry_recovery_shows_a_retry_then_success passed.

6. **Complete Provider Outage Handling (G6)**

   Issue: Unhandled upstream API outages generated confusing internal 500 errors.

   Fix: Controlled HTTP 502 Bad Gateway response with explicit diagnostic payload and attempt history.

   Verification: test_model_failure_reports_one_origin_with_its_attempts passed.

4. Known Boundaries & Pre-Production Deployment Notice

While all automated tests pass, the following operational boundaries are documented for evaluators:

1. **Authentication Boundary:** The system currently relies on synthetic cookie sessions and mock student records. It **does not authenticate student identity**. End-to-end authorization (checking student ownership of specific cases) requires upstream Single Sign-On (SSO) integration before connecting to live Makerere student information systems.
2. **Quota Bounds:** Full live-model evaluation requires an active, funded Google Gemini API tier to avoid HTTP 429 rate limits during multi-case automated evaluation runs.